FINDED: AN INTERACTIVE WEB AND MOBILE CLASSROOM LOCATION AND SCHEDULING FINDER SYSTEM FOR KABANKALAN CATHOLIC COLLEGE

BSIT Capstone Project Concept Paper

INTRODUCTION

Background of the Project

Kabankalan Catholic College (KCC) serves as one of the premier higher educational institutions in Kabankalan City, Western Visayas, offering a diverse array of academic programs across multi-story buildings and expanding campus facilities. As the institution's student population grows and academic schedules become increasingly dynamic, efficient spatial navigation across campus becomes vital for daily academic operations. Currently, freshman students, transferees, working scholars, visiting faculty members, and guest speakers frequently encounter significant difficulties in locating designated classrooms, specialized laboratories, faculty offices, and administrative departments during the initial weeks of each academic semester.

The existing classroom location process relies heavily on static physical campus maps posted at major gates, paper-based schedule slips, and verbal inquiries directed toward security personnel or peer students. This manual approach frequently leads to confusion, congestion in hallways, tardiness during class transitions, and unnecessary administrative strain on campus personnel. Furthermore, ad-hoc room transfers and schedule modifications initiated by academic departments are difficult to disseminate in real time, causing misplaced classes and temporary spatial disruption.

To eliminate these navigation bottlenecks, the proposed project will focus on developing FindED: an interactive web-based and mobile classroom location and scheduling finder system specifically customized for Kabankalan Catholic College. The

system will feature an interactive 2D/3D campus floor layout map, real-time classroom occupancy status tracking, intelligent route navigation, and integrated schedule management. By digitizing spatial mapping and synchronizing room assignments with academic schedules, FindED will provide an intuitive, accessible, and efficient campus navigation pipeline for students, faculty members, and institutional administrators.

Statement of the Problem or Project Objectives

The main objective of this project is to design, develop, and evaluate FindED, an interactive web and mobile campus navigation and classroom location system for Kabankalan Catholic College, to streamline room identification, schedule lookup, and campus spatial orientation.

Specifically, the project aims to accomplish the following technical objectives:

- 1. Develop a interactive campus mapping module that displays detailed floor plans, room designations, building layouts, and key campus facilities of Kabankalan Catholic College.
- 2. Implement an automated room and schedule search engine that enables users to query classroom locations by room code, subject code, instructor name, or section schedule.
- 3. Integrate a real-time room availability status tracker displaying whether classrooms are currently occupied, vacant, or reserved for institutional events.
- 4. Establish role-based access controls for System Administrators, Academic Department Heads, Faculty Members, and Students, complemented by comprehensive audit trails to maintain data integrity.
- 5. Conduct user acceptance testing (UAT) among KCC students, faculty members, and administrative staff to evaluate system usability, navigation accuracy, and operational efficiency.

Significance of the Project

The proposed FindED system will provide direct practical value and technological benefits to the following key stakeholders:

- **Students:** Eliminates campus navigation confusion, minimizes class tardiness, provides immediate access to room assignment updates, and significantly improves the daily academic experience, particularly for freshmen and transferees.
- **Faculty Members and Instructors:** Facilitates seamless identification of assigned lecture halls and laboratories, minimizes instructional delays caused by lost students, and assists ad-hoc room swapping management.
- **Campus Administrators and Registrar Personnel:** Streamlines room utilization tracking, optimizes spatial allocation across campus facilities, and reduces administrative inquiries regarding room assignments.
- **Campus Visitors and Guests:** Offers an accessible digital directory and directional utility for guest lecturers, parents, and event participants navigating the college campus.
- **Future Researchers and IT Students:** Serves as a reference model for localized spatial navigation systems, indoor mapping frameworks, and institutional scheduling software applications.

Alignment with Institutional Research Agenda and Sustainable Development Goals

Institutional Research Agenda Alignment: The proposed project aligns directly with the Institutional Information Technology and Smart Campus Governance research priorities. By introducing digital spatial mapping and computerized schedule management, the project advances the modern technological infrastructure of Kabankalan Catholic College and supports institutional digital transformation.

Sustainable Development Goal Alignment:

- **SDG 4 (Quality Education):** Enhances the educational environment by optimizing learning time, reducing class disruptions caused by navigation delays, and providing equal navigational access to all campus facilities.
- **SDG 9 (Industry, Innovation, and Infrastructure):** Promotes technological innovation and digital campus infrastructure by deploying localized web and mobile application solutions within an educational setting.
- **SDG 11 (Sustainable Cities and Communities):** Contributes to smart campus sustainability by reducing paper consumption associated with printed schedules and static physical orientation signages.

SCOPE AND LIMITATIONS

System Scope & Technical Architecture

The scope and operational boundaries of the FindED system are structured into core system components and technology stack standards as presented in Table 1 below:

Table 1. System Scope & Technical Architecture Specifications

Component	Description & Coverage
Core Modules	Interactive Campus & Floor Map Viewer, Classroom Search Engine, Real-time Room Schedule & Occupancy Directory, Ad-hoc Room Reassignment Portal, and System Security Audit Log.
Technology Stack	Web and mobile responsive architecture utilizing Node.js/ PHP for backend API services, MySQL/PostgreSQL for database management, and HTML5 Canvas/SVG, CSS3,

Component	Description & Coverage
	JavaScript, and Flutter/React Native for interactive map rendering and mobile user interfaces.
Target Users	Kabankalan Catholic College Students, Academic Faculty Members, Department Deans, Campus Security Staff, and System Administrators.

Limitations

To maintain project feasibility within the allotted capstone development period, the following boundaries and constraints are defined:

- 1. The interactive visual mapping will be restricted specifically to the physical campus boundaries and buildings of Kabankalan Catholic College main campus.
- 2. Hardware-based indoor GPS positioning, Bluetooth low-energy (BLE) beacons, and hardware RFID tracking sensors are excluded from the current development phase. Spatial location will be driven by floor plan navigation and room code indices.
- 3. Automated automated integration with third-party external university payment systems or SMS alert gateways is excluded from the current phase and may be recommended for future enhancement.